Exam Date & Time: 21-Nov-2024 (02:30 PM - 05:30 PM)

# MANIPAL ACADEMY OF HIGHER EDUCATION

## DOWNSTREAM PROCESSES- II [BIO 3124]

**Marks: 50** **Duration: 180 mins.**

**Descriptive**

**Answer all the questions.** Section Duration: 180 mins

1A) Generate x-y data for binary system with constant relative volatility of α=2.14 and plot (10 x-data points from 0 to 1). (2)

1B) The following is the vapour pressure data of Methanol-Ethanol at 760 mmHg total pressure. Generate T-x-y data using Raoult's law; also compute relative volatility, α. (4)

| T, $^0$C | $P^0_A$, mmHg | $P^0_B$, mmHg |
|---|---|---|
| 64.7 | 760 | 439 |
| 68 | 860 | 503 |
| 71 | 961 | 568 |
| 74 | 1072 | 640 |
| 75.5 | 1131 | 679 |
| 78.4 | 1253 | 760 |

1C) It is desired to dry the soap from 20% to 4% moisture by placing in a large container of air, whose initial water vapor partial pressure is 16mmHg. The pressure and temperature will be maintained throughout at 1std atm and 50°C. For 1 kg initial wet soap, what is the minimum volume of the container required in $m^3$ ? Vapor pressure of water at $50^0$C is 92.5 mm Hg. R=8.314 J/mol.K. State your assumption. (4)

2A) 10kg of dry soap is placed in a vessel containing 9 $m^3$ moist air whose initial moisture content was found to be 85% RH, to adsorb moisture. The operation is continued till the RH reached a value of 35%. The total pressure and temperature are maintained at 1 std atm and 50°C respectively. What will be the ultimate moisture content of the soap on dry basis? Vapor pressure of water at $50^0$C is 92.5 mm Hg. R=8.314 J/mol K. The total volume of the vessel drops by 5%. (4)

2B) Following is the isothermal adsorption data of a medicine in water solution (x) by an adsorbent (q). Fit Langmuir isotherm for the adsorption. $q=q_0x/(K+x)$ (4)

| q (mg-medicine/$cm^3$ adsorbent) | 5.1 | 6.9 | 7.9 | 9.5 | 10.7 |
|---|---|---|---|---|---|
| x (mg-medicine/Litre solution) | 0.011 | 0.019 | 0.029 | 0.06 | 0.111 |

2C) A colored dye solution is passed through a packed bed of adsorbent. Draw the breakthrough [$C/C_0$ (y) vs elution volume (x)] illustrating the following parameters (i) bed height is varied (ii) flow rate is varied; provide explanation. (2)

3A) A chromatographic separation of a two component samples on a 50 cm column gave the retention times for the solutes A and B as 2.5 and 3.1 minutes with base widths of the two chromatographic peaks being 0.24 and 0.3 cm resp. Calculate the (a) Number of theoretical plates (b) Average plate height and (c) Resolution of the two peaks; $n=16(RT/w_b)^2$; use the definition of resolution for its computation. (2)

3B) Elucidate (i) How does the Spacer arm function in a Dye Ligand Chromatography? (ii) How protein separation is achieved in a Chiral chromatography (iii) How does IMAC (Immobilized Metal Affinity Chromatography) function (iv) Selection criteria for buffers in an Ion Exchange Chromatography (4)

3C) With regard to Size exclusion chromatography (i) plot the relationship between retention volume versus molecular weight of solute (ii) effect of adsorption on $k_d$ (iii) protein shape and size effect on $k_d$ (iv) disadvantages of size exclusion chromatography (4)

4A) How does the Eddy diffusion, Longitudinal diffusion and Mass transfer affect the performance of a chromatographic column? (3)

4B) A batch of saturated $Na_2CO_3$ solution of 100kg is to be prepared at $50^0$ C. The solubility is 4.48 g/1000g water at the temperature. If monohydrate were available, how many kg of water would be required to form the solution? If decahydrate is available, how many kg of salt will be required to form the solution? Mol wt of $Na_2CO_3$=106. (4)

4C) State Meir's theorem. What are the assumptions and drawbacks? (3)

5A) Elucidate the 4 important factors on which crystals habits depend (3)

5B) Draw the drying rate (N) vs moisture content (X) for the following cases (i) the non hygroscopic solid where moisture loss by mass transfer was dominated by boundary layer at the air-solid interface (ii) the partially hygroscopic solid where moisture loss by mass transfer was dominated by diffusion through the solid. Give proper reasons. (3)

5C) The following is the drying rate [kg/(kg.h)] data for a meat sample. Compute cumulative moisture removed as a function of time in minutes. Use average rate for each time interval. (4)

| Time, min | 0 | 5 | 10 | 15 | 20 | 30 | 40 | 50 | 60 | 70 | 80 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Rate , kg/(kg.h) | 0 | 0.25 | 0.45 | 0.6 | 0.7 | 0.8 | 0.85 | 0.82 | 0.8 | 0.79 | 0.75 |

-----End-----